

Research Inspiration / Benchmark Patterns

This project is inspired by how consumer apps and super-app-like platforms use personalization and recommendation to help users discover relevant actions, services, content, or offers.

The goal is not to replicate any large-scale production recommender system. Instead, this project extracts common patterns and adapts them into a small, transparent, and evaluable prototype.

Product / Domain	What they do	Pattern to learn	How this project adapts it
Grab / super-app	Uses real-time personalization and predicted intent to trigger personalized actions such as payment discounts or dining offers.	Predicted intent → personalized action.	Use simulated user intent to recommend cashback campaign, merchant offer, education card, payment shortcut, or re-engagement action.
MoMo / Vietnam fintech app	Public AI/ML hiring signals mention product recommendation, personalization, promotion targeting, eKYC, risk scoring, fraud detection, and financial services.	Personalization and promotion targeting are relevant in Vietnamese fintech/consumer-app environments.	Include actions such as voucher, cashback, payment-related action, loyalty feature, and finance education content.

Shopee / e-commerce	Recommendation systems power personalized shopping experiences; related research explores graph-based item retrieval for recommendation ads.	User behavior graph and item/action retrieval.	MVP uses simpler behavior features and ranking; future work can explore user-action graph recommendation.
DoorDash / food delivery	Homepage recommendation balances exploitation and exploration.	Avoid only recommending what the user already likes; include discovery.	Add exploration bonus and fatigue penalty so the same action is not recommended repeatedly.
YouTube / content recommendation	Uses a two-stage recommendation architecture: candidate generation followed by ranking.	Generate candidates first, then rank them.	First generate eligible next actions, then rank top actions for each user.
Spotify / content personalization	Uses taste-profile signals and gives users some control over how recommendations are shaped.	Explainable and controllable user profile.	Show why a user is assigned a persona and allow simulated feedback such as “show less cashback” or “not interested in finance education.”

**Amazon /
e-commerce**

Uses behavior and intent-aware recommendation to improve relevance and product discovery.

Behavior → intent → recommendation.

Infer user persona/intent from behavior events, then recommend the next best action.

Key design patterns extracted:

1. **Two-stage decisioning**: candidate action generation → action ranking.
2. **User persona / intent modeling**: behavior history becomes an interpretable user profile.
3. **Context-aware recommendation**: time, recency, inactivity, and feature usage affect the recommendation.
4. **Exploration vs exploitation**: recommend relevant actions while still discovering new interests.
5. **Eligibility and safety rules**: not every high-scoring action should be shown.
6. **Explainability and feedback loop**: the system should explain recommendations and learn from user feedback.

These patterns will be implemented in a simplified form using synthetic data, feature engineering, rule-based baseline, optional clustering/ranking model, and evaluation through synthetic scenarios.